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SOCIAL SEPARATION IN INFANT *CEBUS APELLA*: PATTERNS OF BEHAVIORAL AND CORTISOL RESPONSE[1]

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Abstract—34 infant tufted capuchins (*Cebus apella*) were separated from their social groups for a 2-hour period and videotaped in isolation at the ages of 6 months and 1 year. Baseline and 2-hour blood samples were measured for levels of serum cortisol. Compared to homecage baseline levels, passivity, locomotion and vocalizations increased during separation, while self-directed behavior and environmental exploration decreased. Both behavioral and cortisol responses to separation showed individual stability over the 6 month period, although both responses were somewhat attenuated at the later age. There was little correlation between cortisol and behavior during separations. Females vocalized more than did males during separations and showed greater cortisol increases at 6 months of age. The pattern of behavioral response seen in the 2 hours following separation appeared to be more passive than the typical 'protest' response described in many nonhuman primates, and may reflect either the physical circumstances of the separation or a characteristic of species with relaxed social bonds and considerable allomothering available to infants. Published by Elsevier Science Ltd

INTRODUCTION

The study of social separation in nonhuman primates has a long and diverse history. The process of separating infants from mothers or social groups has been useful for examining variation in biobehavioral responses to and recovery from stress in different primate species and environments.[2,3,6–8,12,15,18,19,22,24] Among the lower primates, capuchin species are faced with a somewhat unique developmental challenge. Comparatively altricial at birth, *Cebus* develop at a slower rate than many other species [4] and go through an extended period of maternal dependency and juvenescence. Over this period they develop into highly adaptive, manipulative adults with comparatively large brains that enable them to learn the complex foraging skills necessary for self-sufficiency.[1,21] To date, only one study has examined the effects of social separation in *Cebus*,[9] focusing on physiological responses to separation in adults. Our lab is currently studying individual differences in behavioral tendencies in infant *Cebus apella* and is using a brief separation paradigm to quantify infants' behavioral and cortisol response to this acute form of social stress. The present report describes the patterns of behavior and cortisol response to a 2-hour separation at 6 and 12 months, respectively, as well as the continuity of response from one age to the next.

EXPERIMENTAL PROCEDURES

Subjects

34 infant *Cebus apella* (21 males, 13 females) from 9 social groups housed at the NIH Animal Center in Poolesville, MD, provided data for this study. Groups ranged in size from 4 to 20 animals and included adults and juveniles of both sexes. Most of the groups were housed in indoor cages (0.9×1.8×1.7 m); however, the largest groups were housed either in indoor/outdoor runs (two connecting areas, each 2.5×3×2.5 m) or in runs during fall and winter and in an outdoor corncrib (4.2 m high, 5 m in diameter) during the spring and summer. 22 subjects (16 males, 6 females) were tested at both 6 months and 1 year of age. An additional 9 infants (4 males, 5

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females) were tested only at 6 months, and 3 more were tested only at 1 year (1 male, 2 females). All subjects lived with their mother in a familiar social group from birth.

Procedure

At the ages of 6 months and 1 year, each subject was removed from its social group and a baseline blood sample was drawn. The infant was restrained manually while blood was drawn from its femoral vein. Every effort was made to collect samples in under 5 min in order to minimize the effect of capture stress on cortisol. After blood was drawn, the infant was placed in a 61×61×76 cm cage in a room out of visual and auditory contact with conspecifics. Food (monkey chow) and water were provided in the cage, as was a perch and a hanging length of plastic chain approximately 36 cm long. The infant was left alone in this room for 2 h, during which time a video camera recorded behavior and vocalizations. At the end of the 2-hour period the infant was removed and another blood sample was drawn; the infant was then returned to its home cage. Serum cortisol values were obtained by radioimmunoassay, performed by Corning Hazelton Laboratories, Vienna, Va.

Behavioral scoring

The first and last 10 minutes of the videotape session were scored for the behaviors listed in Table 1: passive, locomotion, stereotypy, environmental exploration, huddle, self-directed behavior and vocalization. Vocalization was scored as frequency; all other behaviors were scored as duration measures. (It should be noted that all of the self-directed behavior observed consisted of normal self-grooming activities such as scratching or urine-washing; there was no evidence of self-mouthing or self-clasping such as is seen under stress in macaques reared in some abnormal social environments.[23] These separation sessions were scored with a keyboard event recorder and uploaded to a PC for summary and analysis.

In addition, for 25 of the subjects at 6 months and 20 of the subjects at 1 year, 2–3 10-min videotape sessions in the week preceding separation were available for analysis. These videotapes were scored for the occurrence of the behaviors listed above, to provide pre-separation measures of these behaviors under baseline conditions.

Data analysis

Variation in behavior and cortisol values (baseline, 2-hour peak, and change from baseline to peak (Δ cortisol)) were analyzed via mixed design ANOVAs. Sex served as a between-subject variable, and age (6 months vs 1 year) and time block within each separation (first and last 10 min for behavior, baseline vs 2-hour peak for cortisol) served as within subject variables in these analyses. Behavioral scores in the first and last 10 min were then averaged to provide a mean value for each separation, used in further analyses. Differences between pre-separation behavior levels and separation scores were examined using analogous ANOVAs (sex as between-subject variable, age and pre-separation vs separation as within-subject variables). An alpha level of 0.05 was selected in determining significance of results of ANOVAs.

Behavior and cortisol scores during the two separations were correlated to examine individual consistency across the 6-month time span. Pearson correlations between cortisol and behavior values were examined to determine the relationship between behavioral and physiological re-

Table 1. Behavioral definitions

Passive — sitting, standing, or lying motionless, not engaged in any other scored activity. (When scored in pre-separation, scored only if alone (>15 cm from another animal) OR if riding dorsally on mother and not engaged in any other activity)
Locomotion — movement resulting in displacement of more than 15 cm from original position
Stereotypy — repetitive locomotive pattern of more than 3 cycles; idiosyncratic behavior such as head-tossing
Environmental exploration — tactile, oral, or olfactory exploration of external environment (cage, food, etc.)
Huddle — sitting passive with head lower than point of shoulders
Self-directed — self-mouthing, self-clasping, self-grooming (including urine washing), scratching
Vocalization — any audible oral sound

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sponses to the stress of separation at each age. Because of the exploratory and descriptive nature of the correlational analyses, an alpha level of $p < 0.10$ was adopted in examining those results.

RESULTS

When isolated in a small cage away from conspecifics at 6 months, capuchin infants showed marked increases in passivity, locomotion and vocalizations from baseline levels and decreases in self-directed behavior and environmental exploration. Cortisol levels two hours after separation at 6 months reached levels 9–10 times higher than baseline. During a second separation 6 months later, the behavioral and physiological responses were similar but somewhat attenuated.

Pre-separation vs separation behavior. Compared to homecage behavior in the week before the separations at 6 months of age, passive, locomotion and vocalization all increased when animals were separated, while environmental exploration and self-directed behavior decreased (Table 2). A similar pattern was seen at 1 year of age, except that locomotion did not change significantly at this age and stereotypy decreased from preseparation to separation levels.

Within-session changes. At the age of 6 months, self-directed behavior and stereotypy were significantly higher in the last 10 minutes of the separation session than in the first 10 min; over that time period vocalizations decreased (Fig. 1). At the age of 1 year, self-directed behavior, huddling and stereotypy increased from the first to last part of the session; environmental exploration also increased in these subjects while passive and vocalization decreased.

Stability of separation behaviors across age. Passive, locomotion, stereotypy, environmental exploration and vocalization all showed individual stability from 6 months to 1 year of age ($r = 0.366$–0.695, $p < 0.10$, Fig. 2). The mean frequency of vocalization decreased from 6 months to 1 year of age, while time spent in passive increased at the later age (Table 2, Fig. 3(a)). Pre-separation levels of behavior at 6 months and 1 year were not significantly correlated. Pre-separation levels of locomotion ($F = 15.01$, 1,14 df, $p = 0.001$) and stereotypy ($F = 4.578$, 1,14 df, $p < 0.05$) increased from 6 months to 1 year (Table 2).

Stability of cortisol. Both peak and Δ cortisol scores were lower at 1 year than at 6 months (Fig. 3(b)). There was moderate individual stability for peak ($r = 0.389$, $p < 0.10$) and Δ cortisol scores ($r = 0.568$, $p < 0.05$) from 6 months to 1 year, but not for baseline measures (Table 3, Fig. 4).

Correlations between cortisol and behavior. The only overall relationship observed between cortisol and behavior during separations was a negative correlation between huddling and Δ cortisol scores at 6 months (Table 4).

Table 2. Comparison of preseparation and separation behavior levels. Means and standard errors of behaviors scored in the week preceding separations and during the 1st and last 10 minutes of separations. PA = passive, LO = locomotion, ST = stereotypy, HU = huddle, SE = self-directed, EE = environmental exploration, VO = vocalizations. Vocalizations scored as frequency; all other measures expressed as duration in seconds/10 min period. Asterisks indicate significant ($p < 0.05$) difference between pre-separation and separation behavior levels

	PA	LO	ST	HU	SE	EE	VO
6 months							
Presep	107.7*	97.2*	0.0	0.0	41.7*	206.3*	9.4*
sem	31.2	10.0	0.0	0.0	7.6	20.3	5.6
Sep	280.8	164.1	4.5	13.0	7.4	161.2	121.1
sem	21.8	18.1	2.5	11.8	1.7	15.1	22.5
1 year							
Presep	53.0*	135.6	28.0	0.3*	27.3*	209.2*	4.3*
sem	9.6	13.4	13.8	0.3	3.4	17.6	1.2
Sep	342.1	129.6	5.2	51.0	3.5	96.1	46.7
sem	25.4	24.0	2.0	13.5	1.2	17.0	18.6

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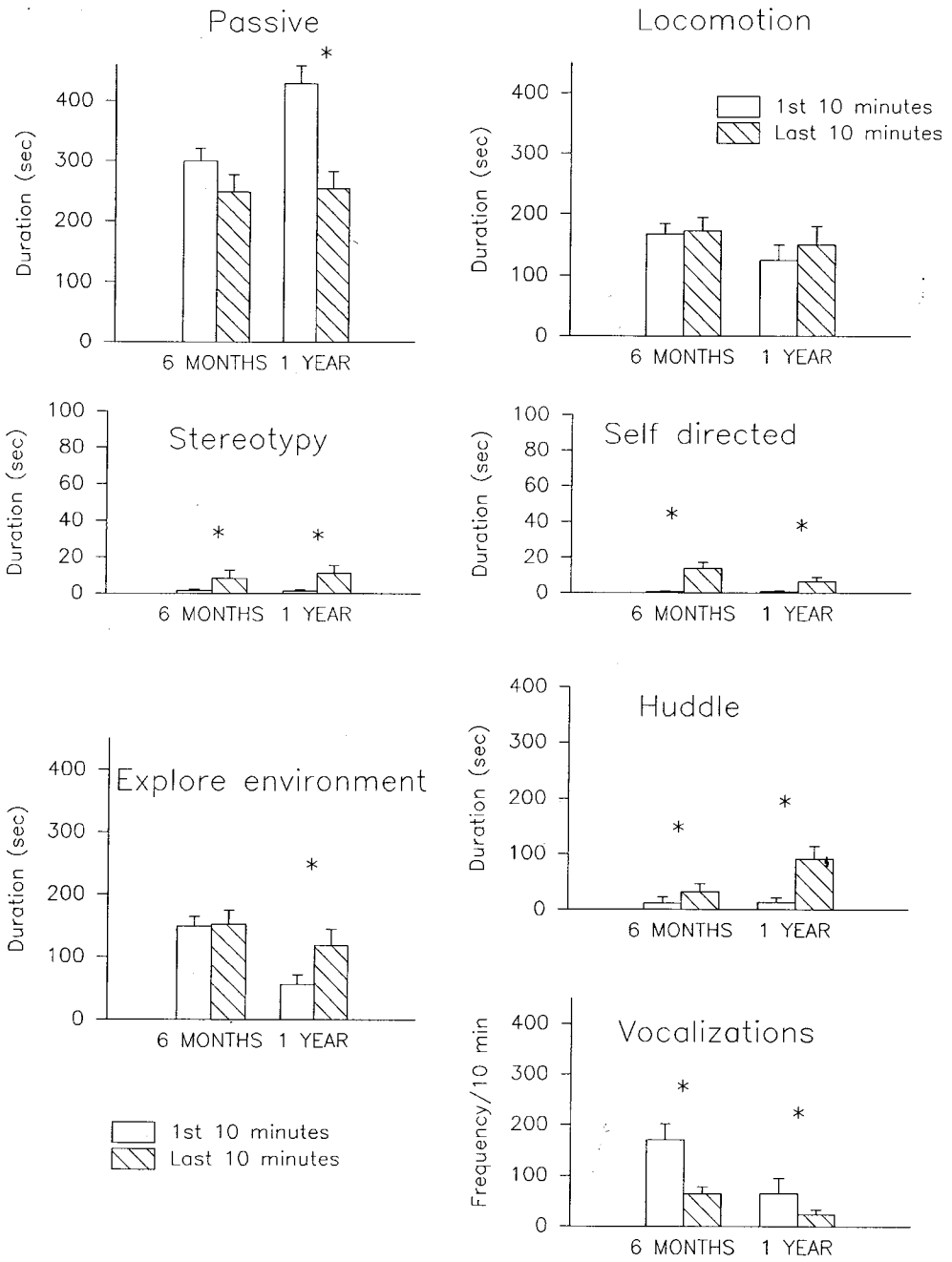


Fig. 1. Behavior during 1st and last 10 min of separation sessions. Asterisks indicate significant mean differences ($p < 0.05$). Vertical bars represent S.E.M.

Sex differences

Behaviorally, the only sex difference during separations was in the frequency of vocalizations: females vocalized more than males, especially in the 1st 10 min of separation sessions (Fig. 5(a)). Even at 1 year of age, when vocals had decreased from levels seen at 6 months, females' frequencies of calls during the 1st 10 min of separation were still extremely high. Males' vocalizations at 1 year did not differ from pre-separation to separation values (5.0 calls/10 min vs 17.1), whereas females' vocals increased significantly at that age (from 2.7 during the week before separation to 115.7 calls/10 min during separation ($F = 6.671$, 1,14 df, $p = 0.019$).

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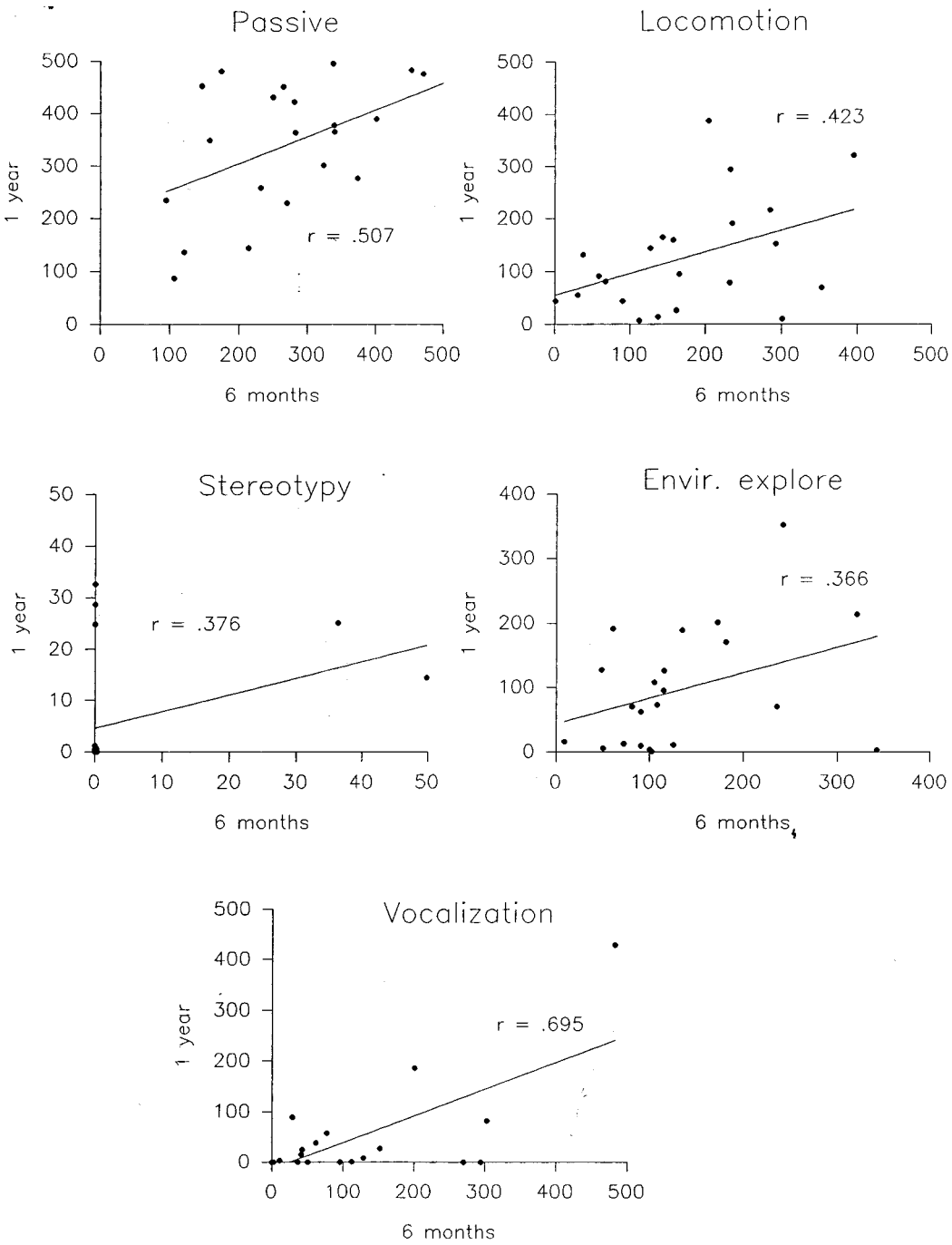


Fig. 2. Pearson correlations of behaviors from 6 months to 1 year. Vocalization is scored as frequency; other behaviors scored as duration measures.

Females showed higher peak and Δ cortisol levels at 6 months than did males (Fig. 5(b)). The overall decrease in cortisol from 6 months to 1 year described above was largely attributable to significant decreases in females from extremely high levels at 6 months.

DISCUSSION

At 6 months and 1 year of age tufted capuchin infants were more passive, huddled more, vocalized more, and explored less when separated than when in their homecage environment. 6-

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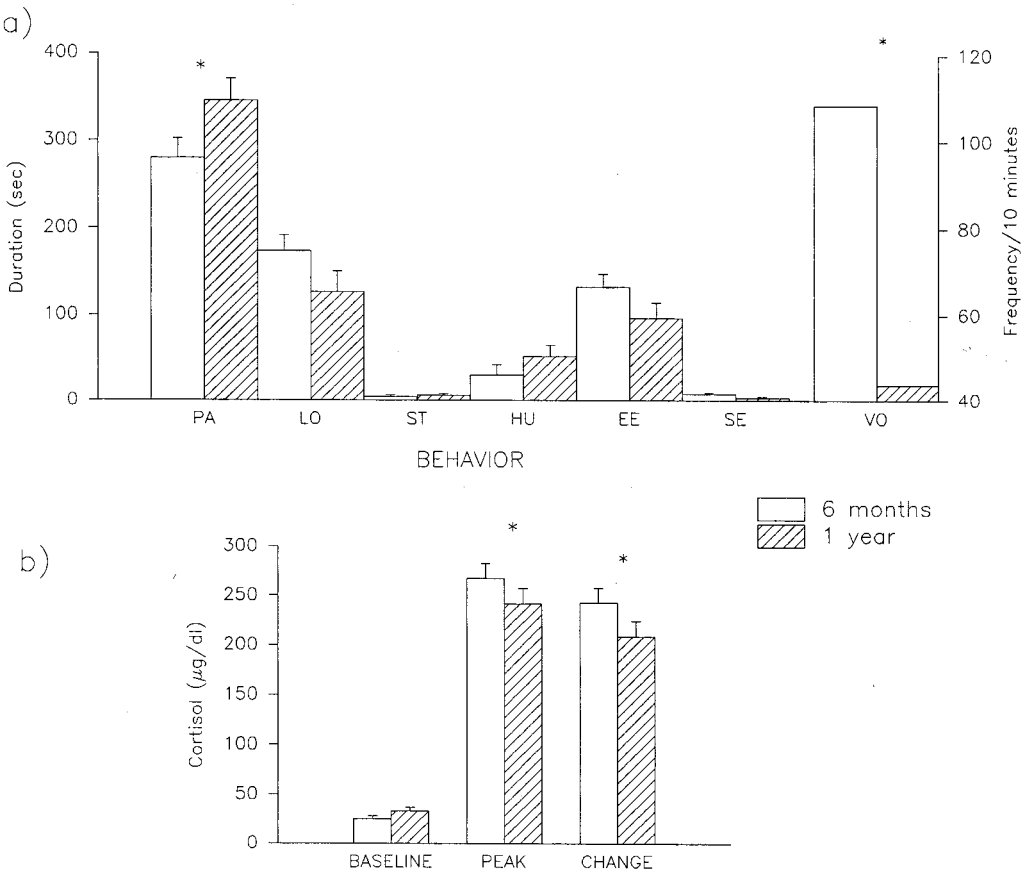


Fig. 3. (a) Behavioral scores during 6 month and 1 year separations. Asterisks denote significant mean differences ($p < 0.05$). PA = passive, LO = locomotion, ST = Stereotypy, HU = huddle, SE = Self-directed, VO = vocalizations. Vocalizations are scored as frequency; all other behaviors scored as duration measures. (b) Cortisol levels during 6 month and 1 year separations. Asterisks denote significant mean difference ($p < 0.05$). Vertical bars represent S.E.M.

month-olds also showed elevated locomotion during separations, perhaps reflecting a lower level of homecage locomotion in younger infants that were still occasionally riding on their mothers. This pattern of response seems somewhat different from the typical 'protest' response to separation (high levels of locomotion or activity) often cited in species such as rhesus macaques.[3,14,19] The 'protest' phase of separation is thought of as an immediate adaptive response by an infant out of contact with its caretaker, with increased activity and vocalizations serving to increase the probability of reunion.[17] At both 6 months and 1 year, the majority of time in the 2-hour separations in this study was spent in passive behavior, with only about 1/4 of the time spent in locomotion.

Separation responses may vary widely according to species, separation paradigm, rearing experience, etc.[3,11,13] Initial results from this study suggest that *Cebus* infants that are isolated from conspecifics for short periods respond for the most part with decreased activity, although

Table 3. Pearson correlations of cortisol values. Base6, Peak6, ΔCort6 = 6 month values. Base1, Peak1, ΔCort1 = 1 year values. $N = 21$, underline = $p < 0.05$, bold = $p < 0.10$

	Base6	Peak6	ΔCort6	Base1	Peak1
Peak6	**0.391**				
ΔCort6	0.246	0.988			
Base1	–0.150	–0.344	–0.337		
Peak1	–0.118	**0.388**	**0.428**	**0.384**	
ΔCort1	–0.080	0.527	0.568	0.098	0.956

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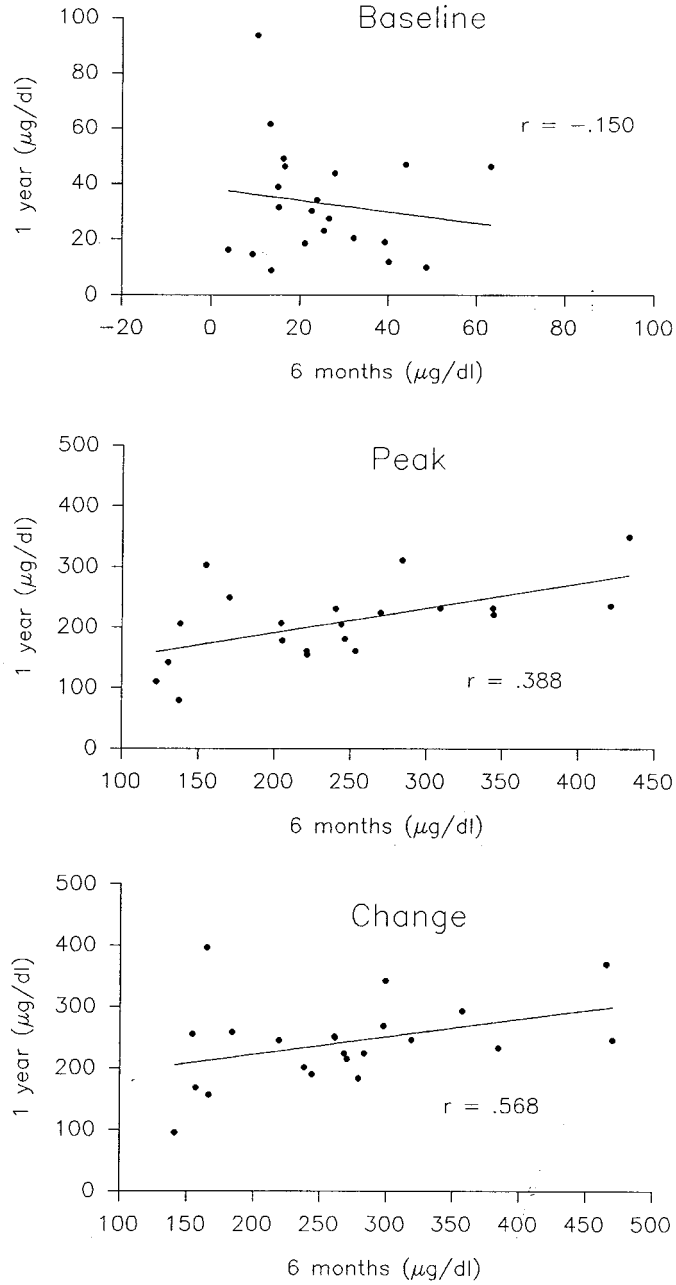


Fig. 4. Pearson correlations of cortisol from 6 months to 1 year.

Table 4. Pearson correlations of cortisol and behavior during separations. $N = 30$ at 6 months, 25 at 1 year, underline = $p < 0.10$

	6 months			1 year		
	Base	Peak	Δcort	Base	Peak	Δcort
Passive	−0.062	−0.024	−0.012	0.056	−0.089	−0.108
Loco	−0.100	0.165	0.175	−0.012	0.164	0.174
Stereo	0.052	0.171	0.168	−0.054	−0.140	−0.133
Huddle	0.171	−0.294	−0.345	0.037	0.083	0.077
EnvExp	−0.031	0.073	0.083	−0.106	−0.183	−0.164
Self	−0.031	−0.009	−0.003	−0.190	0.257	0.317
Vocal	−0.162	0.104	0.143	−0.031	−0.033	−0.025

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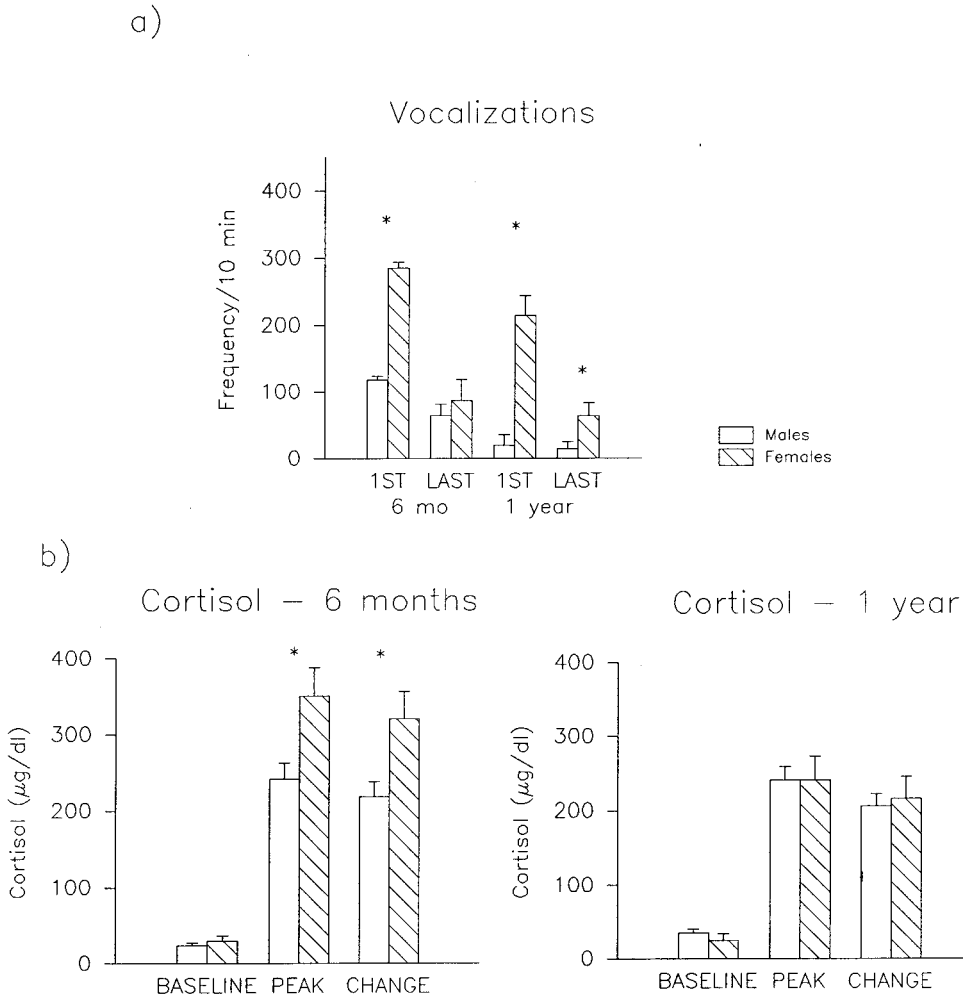


Fig. 5. (a) Sex differences in vocalizations during 1st and last 10 min of separations at 6 months and 1 year. (b) Sex differences in cortisol levels during 6 month and 1 year separations. Asterisks denote significant mean differences ($p < 0.05$). Vertical bars represent S.E.M.

vocalizations do increase significantly above baseline levels. These results are similar to those reported by Levine *et al.* [11], in that squirrel monkeys that were separated away from conspecifics were much less active than those separated in the visual and auditory presence of mothers or other group members. In the latter study, however, baseline behavior levels were not available, so it is unknown how separation behavior levels compared to normative ones in the homecage.

Nonhuman primate infants of species characterized by relatively relaxed social networks, with alternate caretakers available, may also show diminished separation responses under some circumstances.[16] Tufted capuchins fall into this category as well, with infants often receiving considerable allomothering and attention from other group members, even more so than in the closely related squirrel monkey.[5]

Over the course of a 2-hour separation, 6-month-olds showed increases in their huddling and self-directed behavior and decreases in their calling behavior. One-year-olds showed increased exploration, huddling and self-directed behavior over that time, while passivity decreased from the beginning to the end of the 2-hour session, as did frequency of vocalizations. These behavioral changes may represent infants' attempts to cope over the course of a separation session and may reflect an increased repertoire of coping responses available to older infants. For example, only 13% of animals engaged in stereotypy during separations at 6 months, whereas 24% of the subjects did so at 1 year of age. Similarly, 19% of subjects spent time in a huddled posture at 6 months, whereas 52% of subjects did so at 1 year.

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Both behavior and cortisol levels showed individual stability from 6 months to 1 year of age. However, infants appeared to be less active during separation at the later age. They showed less locomotion and vocalization, along with nonsignificant decreases in exploration and increases in passivity and huddling. Along with the increased number of animals that engaged in huddling, as described above, fewer animals vocalized at the older age (84% of 6-month-olds vocalized during separations as opposed to 52% at 1 year). These changes suggest an attenuated response to separation by older animals. It is not known whether this diminished response is a result of physical maturation or an experiential effect of having undergone a similar separation 6 months previously. The stability of individual response patterns and changes over subsequent separations are similar to results of longitudinal studies of rhesus monkeys undergoing repeated separations from infancy to adulthood (e.g., J. D. Higley, 1986, unpublished doctoral dissertation).

Along with a general decrease in behavioral activation was a decrease in cortisol response from 6 months to 1 year. These initial results suggest a possible association between cortisol response and behavioral activity in general. There was little relationship, however, between specific behavioral and cortisol measures at either age over all subjects beyond the one negative correlation between cortisol and huddling at 6 months. By 1 year of age, although huddling was more prevalent, this correlation was no longer evident.

Sex differences

Kling *et al.* [9] reported elevated fecal cortisol levels in adult Cebus after 3–5 d of social separation, with females tending to show a greater response than males. In the present study, 6-month-old female capuchins showed higher cortisol responses to a 2-hour separation than did males; however, the sex difference was not evident at 1 year. The only sex difference in behavioral scores was a greater frequency of vocalizations for females than males, even during the first part of the separation at 1 year. Little mention is made of sex differences in the primate separation literature, although there are a few isolated reports of greater disturbance in males than in females.[10,20] The uneven ratio of males to females in this study makes interpretation of observed sex differences problematic. However, in species such as macaques and capuchins, where males eventually disperse and females remain in their natal groups, it would seem ultimately more detrimental for females to be separated from their groups than males and thus selection pressure might be expected to produce a greater response to such separation in females.

In summary, the pattern of behavioral response seen in *Cebus* infants following a 2-hour social separation appeared to be more passive then the typical 'protest' response described in many studies of nonhuman primates. It is not yet known whether this response is related to the physical circumstances of the separation or to a viable species difference. Further study on the impact of life stressors on behavior and development in the capuchin will aid in clarification of these results.

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